

officer, Velmenni R&D Lab, says, “The visible light spectrum is unlicensed and, hence, is free to use for Li-Fi services. Moreover, Li-Fi enables an evenly distributed network. In certain router-based Wi-Fi setups in enterprises, when antennae are facing towards a specific direction, receivers in that direction get better signal strength compared to others. On the other hand, Li-Fi signals remain consistent throughout a workspace, since luminaires are usually installed in regular intervals.”

Real-world applications

Li-Fi is mostly undergoing R&D at various stages and steadily picking up commercial recognition. Philips Lighting, the first major lighting company to offer Li-Fi-enabled luminaires, has partnered with various government and private bodies to deploy Li-Fi commercially at pilot levels in India. In fact, a substantial portion of its office network utilises Li-Fi.

Philips Lighting’s Li-Fi-enabled LED office lightings for commercial environment mainly come in the form of downlights or 2x2 luminaire with speeds of 30Mbps—sufficient to stream high-quality media files or transfer large volumes of files.

Education and Research Network (ERNET), government of India, has set up a Li-Fi test bed to test the capabilities of Li-Fi in various indoor and outdoor applications. Initially funded internally by ERNET, the government has partnered with IIT-Madras and Philips Lighting to carry out the two-year (2017-2019) planned R&D project.

Dr Neena Pahuja, director general, ERNET, says, “As this technology can provide high speed connectivity, it can be extensively used for applications such as video-on-demand, virtual reality (VR) applications and robotic surgeries in the future. Also, in places where Wi-Fi gets impacted because of interference—for example, radiology area in hospitals and undersea area—Li-Fi can be used for data transmission.”

Current limitations

Solanki mentions that lack of a proper standard is a major reason why Li-Fi

has not yet picked up full pace. He says, “Existing standards like IEEE 802.15.7 (for visible light communication) are outdated and do not define many aspects of Li-Fi. For large-scale adoption, standardisation is essential. Once the standards are in place, Li-Fi integration will become popular in lightings as well as smart devices like smartphones.”

Experts from the connectivity domain and R&D bodies like ERNET are working together to create standards for Li-Fi.

Li-Fi does not have any specific cost points as of now and is more requirement-oriented. Initial investments are higher than a regular Wi-Fi network setup. However, as Li-Fi becomes more popular, prices will reduce. Overall, the technology, at the moment, is mostly restricted to large businesses and government departments.

The way forward

Li-Fi speeds will increase and the technology will become more stable going forwards. For instance, Philips Lighting’s Li-Fi solution will improve, to deliver speeds of 60Mbps by early 2019, and Li-Fi support will be available in tubular lighting form factors as well. Some Li-Fi researches have been able to touch speeds up to 10Gbps, indicating great potential for the technology.

With the right design and miniaturisation in place, in a matter of a few years, Li-Fi will penetrate the small- and mid-size business and consumer segments. Solanki says that this will be put into motion once the standards are in place. “Usually, this takes at least two years,” he adds.

Dr Pahuja adds, “We will see a large number of use cases in smart cities. For instance, a smart pole that can be remotely controlled for further saving of energy, get traffic information via connected cameras and act as a hotspot, as required.”

Joshi concludes, “The visible light spectrum remains an untapped resource with a large bandwidth, making Li-Fi the connectivity technology of tomorrow.”

—Paromik Chakraborty,
technical journalist, EY

Smart Technology To Improve Workforce Safety

➤ Wearable technology and collaborative machines are some of the most reliable and cost-effective safety solutions, especially for safekeeping on-field workforce and resources



Tata Communication's smart watch for industrial safety

Occupational hazards are governed by harsh work environments and lack of safety measures. Globally, more than 2.78 million workers fall victim to occupational deaths. More than 300 million labourers suffer non-fatal injuries at their workplaces every year. This can potentially alter their lifestyles, health and working capabilities permanently.

Apart from the distress of employee risk and lost man-hours, companies also lose a huge amount of money through compensations and damage control—studies imply an average of four per cent GDP loss for countries through employee injuries.

Technologies like the Internet of Things (IoT) and artificial intelligence (AI) can lead industries to substantially reduce workplace mishaps by helping workers take sufficient precaution. These can save many lives and huge financial losses.